RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019-2020 (Odd Semester)

Innovative Practices Description for Unit-I

Degree, Semester& Branch: VII Semester B.E. ECE A

Course Code & Title: EC6013 Advanced Microprocessors and Microcontrollers

Name of the Faculty member: Mr.B.Kannan

Name of the Topic: Pentium Architecture, Addressing Modes and Pentium Programming

Name of the Innovative Practice: Zero Minute Speech, Animation video and MASM Simulation Tool

Date & Duration: 06.07.2019 (10 Minutes), 10.07.2019 (15 Minutes) & 17.07.2019 (15 Minutes)

Description:

(i) Zero Minute Speech:

- Zero Minute Speech will leads to express basics of the particular topics by different students.
- The important topics were allotted to set of students and they will give speech on the topic with doubt clarification.

(ii) Animation video

- Use of audio-visual aids help in maintaining discipline in the class since all the students' attention are focused in learning. This interactive session also develops critical thinking and reasoning that are important components of the teaching-learning process.
- Students learn when they are motivated and curious about something. Traditional verbal instructions can be boring and painful for students. However, use of audio-visual provides intrinsic motivation to students by peaking their curiosity and stimulating their interests in the subjects.

(iii) Simulation Tool:

- Simulation software is based on the process of modelling a real phenomenon with a set of mathematical formulas. It is, essentially, a program that allows the user to observe an operation through simulation without actually performing that operation.
- A primary advantage of simulators is that they are able to provide users with practical feedback when designing real world systems. This allows the designer to determine the correctness and efficiency of a design before the system is actually constructed.

Goals (Learning Outcomes):

- The students will be able to analyse basic concepts of Pentium processor
- The students can be acquired better knowledge about addressing modes of Pentium Processor.
- The students can be program the Pentium processor using MASM Tool.

Use of appropriate method:

Justification for choosing the following Activities:

- **Zero Minute Speech:** Leads to express basics of the particular topics by different students.
- **Audio & Video Tools:** According to the Webster dictionary, audio-visual aids are defined as "training or educational materials directed at both the senses of hearing and the sense. Audio-visual provides opportunities for effective communication between teacher and students in learning.
- **Simulation Tool:** It allows the students to create their own solution for the problem and they will get the real time result.

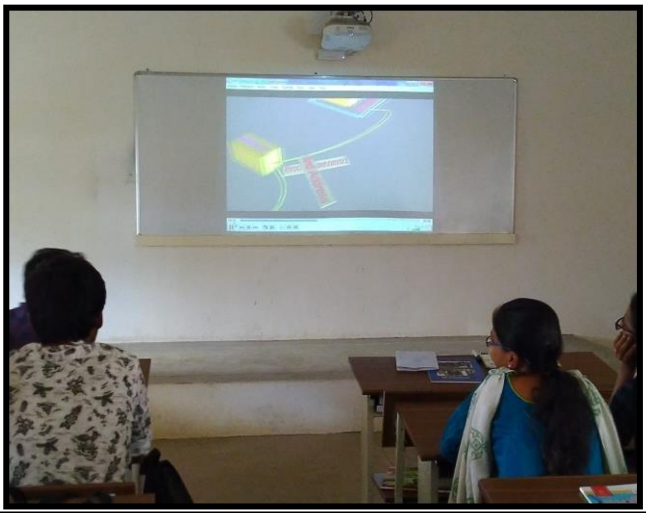
Effective Presentation: (Implementation (Plan & Execution) with Proof):

- **Zero Minute Speech:** I choose this tool because it does not need a big preparation so I can involve the slow-learners also.
- **Animation:** Sometimes audio video tool were lengthy session, so I choose the video clips which more essential to the students.
- **Simulation Tool:** The students completed their activity within the planned time because the procedures are simple.

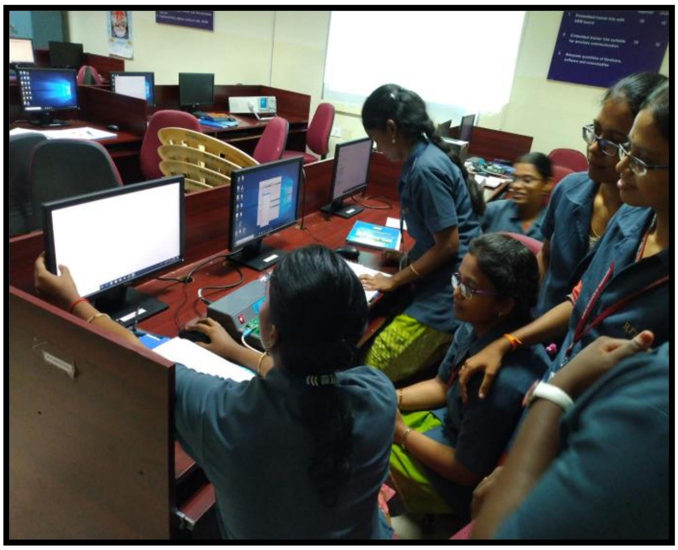
UNIT I

Zero Minute Speech for Pentium Architecture on 06.07.2019

Animation video for Addressing Modes operation on 10.07.2019



Pentium Programming Using MASM Simulation Tool on 17.07.2019



Significance of Results: (Assessment of Effectiveness/Success of the Activity):

- **Zero Minute Speech:** Students delivered the topics with examples about the various blocks of Pentium architecture.
- **Animation:** This technique helps to give lively working of how the processor gets data for the instructions from memory using different addressing techniques.
- **Simulation Tool:** This activity gives the real time solution for their coding.

Changes required for future (if required):

- Need to plan extra hour for simulation tool activity.
- Need to give chance to all students for zero minute's speech.

Reflective Critique:

Challenges:

- **Zero Minute Speech:** Not able to give chance to all students during zero minute speech practice.
- **Animation:** Sometimes there is no audio in animation so faculty member need to explain the step by step working.
- **Simulation Tool:** Few students were not able to follow the procedures in the software.

Benefits:

- **Zero Minute Speech:** Few students share their ideas and information with classmates and some students interacted with other students to clarify the doubts.
- **Animation:**
  - The animation is attractive, It is useful when quickly getting and holding the audience's attention
  - The animation can show the imagined objects in the motion, It is ideal for demonstrating processes
  - Interactive learning with live-action animation keeps learners interested and reinforces skills
- **Simulation Tool:** One of the primary advantages of simulators is that they are able to provide users with practical feedback when designing real world systems.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	1	1			1	1	1	1	1

CO1: The students will be able to describe the operation of Pentium processor and programming concepts.

References:

- *https://www.presentationblogger.com/3-simple-steps-how-to-do-a-speech-on-any-topic-with-zero-preparation/*
- *James L. Antonakos, " The Pentium Microprocessor," Pearson Education, 1997.*
- *https://www.youtube.com/watch?v=LUvN70k5Cpo*
- *http://www.visualmasm.com/*

RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019-2020 (Odd Semester)

Innovative Practices Description for Unit-II

Degree, Semester& Branch: VII Semester B.E. ECE A
Course Code & Title: EC6013 Advanced Microprocessors and Microcontrollers
Name of the Faculty member: Mr.B.Kannan
Name of the Topic: Pentium Architecture, Addressing Modes and Pentium Programming
Name of the Innovative Practice: Audio & Video Tools, Minute Paper and Simulation Tool
Date & Duration: 23.07.2019 (30 Minutes), 26.07.2019 (10 Minutes) & 13.08.2019 (25 Minutes)

Description:

(i) Audio & Video Tools:

- According to the Webster dictionary, audio-visual aids are defined as "training or educational materials directed at both the senses of hearing and the sense. Audio-visual provides opportunities for effective communication between teacher and students in learning.
- Use of audio-visual aids help in maintaining discipline in the class since all the students' attention are focused in learning. This interactive session also develops critical thinking and reasoning that are important components of the teaching-learning process.
- Students learn when they are motivated and curious about something. Traditional verbal instructions can be boring and painful for students. However, use of audio-visual provides intrinsic motivation to students by peaking their curiosity and stimulating their interests in the subjects.

(ii) Minute Paper:

A “one-minute paper” may be defined as a very short, in-class writing activity (taking one-minute or less to complete) in response to an instructor-posed question, which prompts students to reflect on the day’s lesson and provides the instructor with useful feedback.

(iii) Simulation Tool:

- Simulation software is based on the process of modeling a real phenomenon with a set of mathematical formulas. It is, essentially, a program that allows the user to observe an operation through simulation without actually performing that operation.
- A primary advantage of simulators is that they are able to provide users with practical feedback when designing real world systems. This allows the designer to determine the correctness and efficiency of a design before the system is actually constructed.

Goals (Learning Outcomes):

- The students will be able to analyse basic blocks of ARM processor.
- The students will be able to describe the stages of ARM pipeline.
- The students will be able to develop a simple program using IAR development tools.

Use of appropriate method:

Justification for choosing the following Activities:

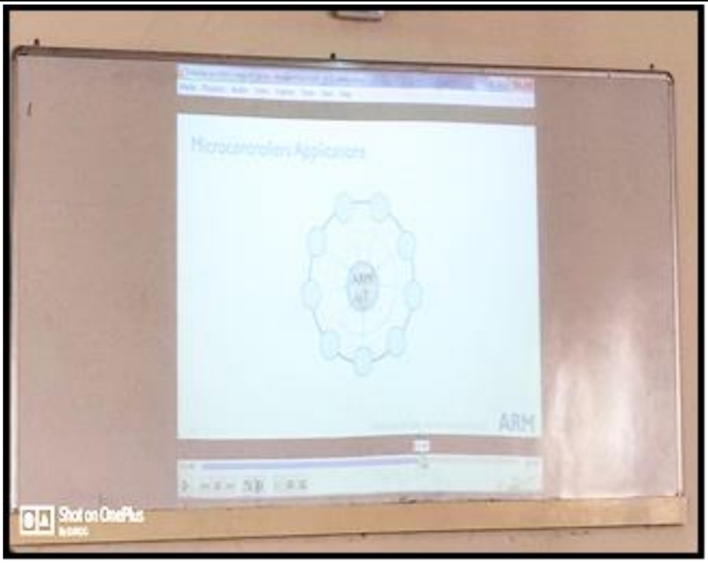
- **Audio & Video Tools:** According to the Webster dictionary, audio-visual aids are defined as "training or educational materials directed at both the senses of hearing and the sense. Audio-visual provides opportunities for effective communication between teacher and students in learning.
- **Minute Paper:** The great advantage of Minute Papers is that they provide manageable amounts of timely and useful feedback for a minimal investment of time and energy.
- **Simulation Tool:** It allows the students to create their own solution for the problem and they will get the real time result.

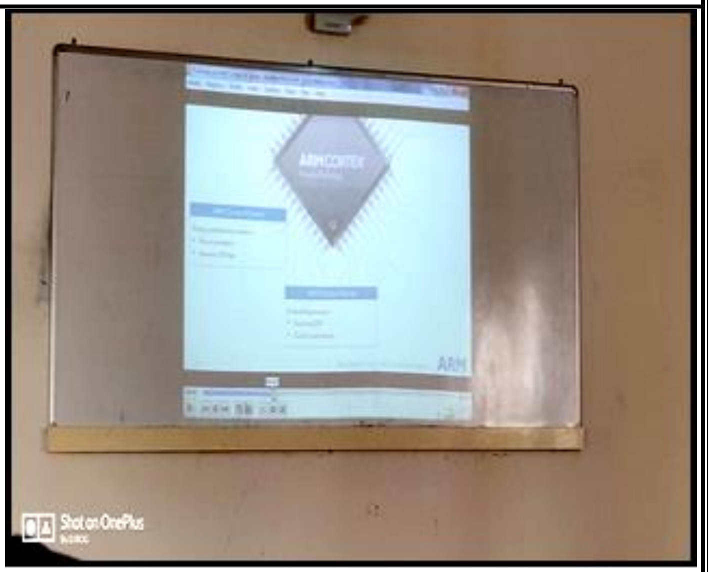
Effective Presentation: (Implementation (Plan & Execution) with Proof):

- **Audio & Video Tools:** I planned to play the animation video for 30 minutes but it took 10 more minutes to complete because in between interaction about few topics with students.
- **Minute Paper:** I went with cut paper to execute the activity, due to on spot topic few students were not able to finish the activity.
- **Simulation Tool:** With new workbench software for the activity took much time compare to what I have already planned but the students got answer for most of the programs.

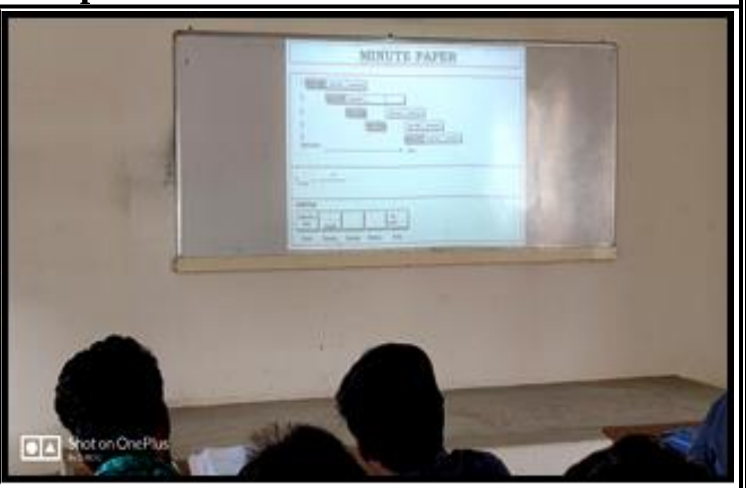
Innovative Teaching Method Execution

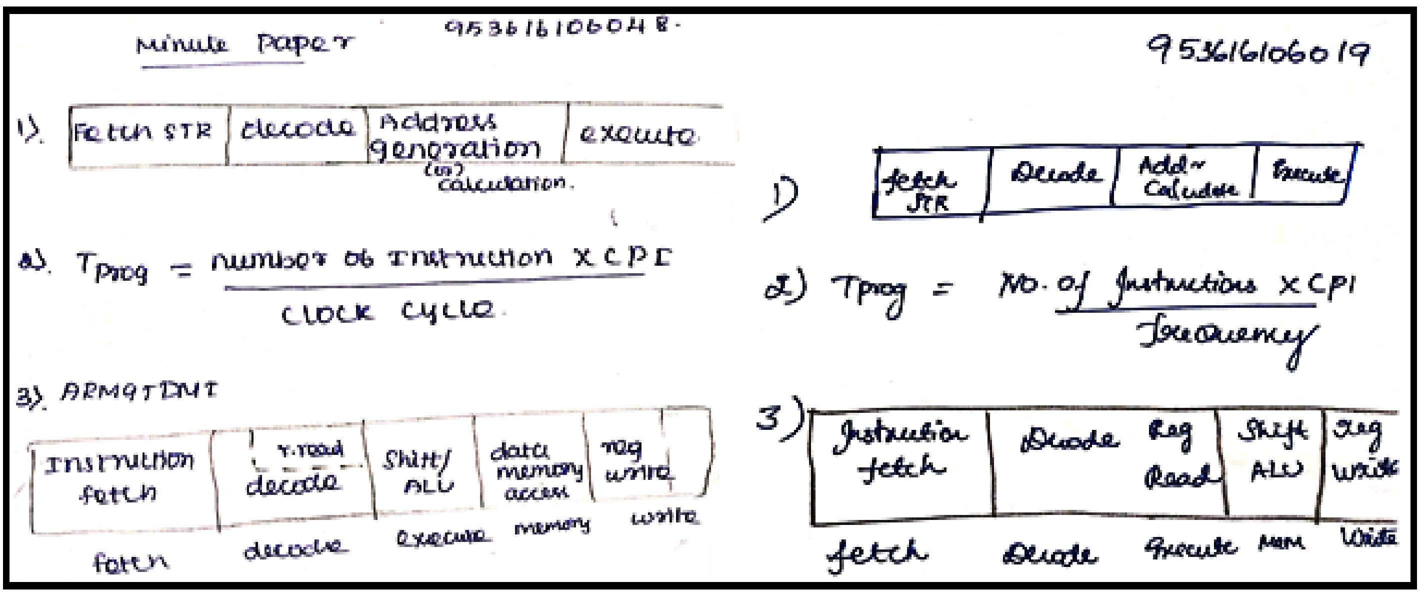
Audio & Video Tools for ARM Architectural Inheritance on 23.07.2019





Minute Paper for ARM Pipeline on 26.07.2019





Minute Paper 95361610604 8.

1). | Fetch STR | decode | Address generation (or) calculation. | execute |

2). $T_{prog} = \frac{\text{number of Instruction} \times CPI}{\text{clock cycle}}$

3). ARM9TDMI

Instruction fetch	r.read decode	Shift/ALU	data memory access	reg write
fetch	decode	execute	memory	write

9536161060 19

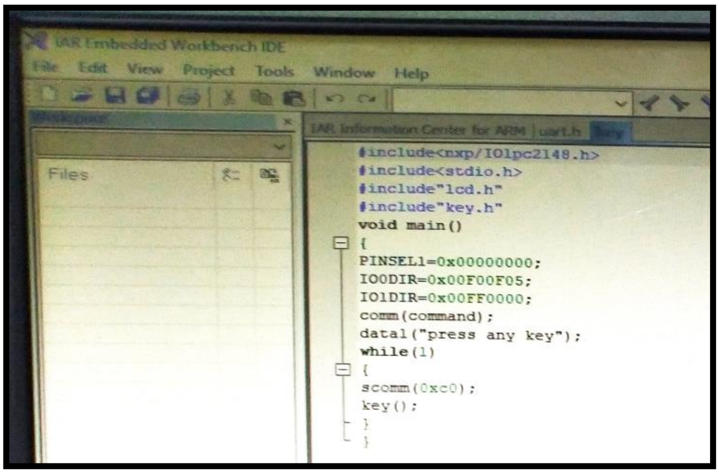
1) | fetch STR | Decode | Add~ Calculate | Execute |

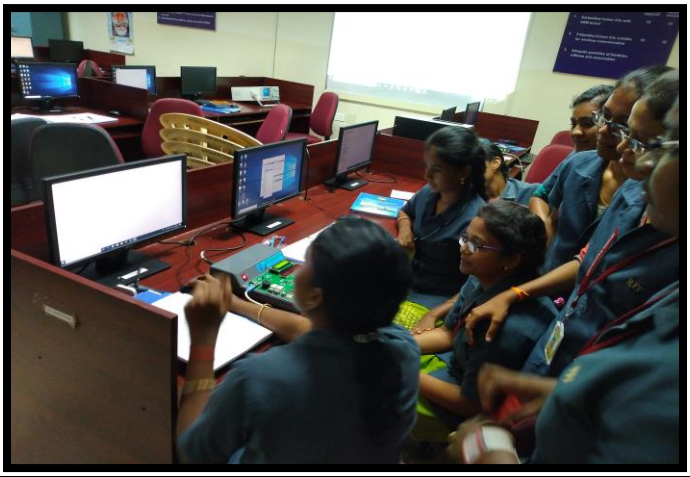
2) $T_{prog} = \frac{\text{No. of Instructions} \times CPI}{\text{Frequency}}$

3)

Instruction fetch	Decode	Reg Read	Shift ALU	Reg write
fetch	Decode	Execute	MEM	Write

ARM C Programming Using IAR Integrated Development tool on 13.08.2019





Significance of Results: (Assessment of Effectiveness/Success of the Activity):

- **Audio & Video Tools:** The animation video gives the clear working and function of each block so students understand the function of Pentium processor.
- **Minute Paper:** Helps the students to recall the examples for each addressing modes.
- **Simulation Tool:** It provides real time output and allows the students to correct their mistake until the final result.

Changes required for future (if required):

- Need to give brief introduction on procedures to follow in the simulation tools to execute the programs effectively.

Reflective Critique:

Challenges:

- **Audio & Video Tools:** Sometimes audio video tool were lengthy session. Student Distractions, visual aids are more of a distraction if used throughout the entire presentation versus during key points.
- **Simulation Tool:** It provides users with practical feedback.

Benefits:

- **Audio & Video Tools:**
  1. Make learning process more effective and conceptual.
  2. Its helps to grab the attention of students.
- **Minute Paper:**
  1. Provide a “conceptual bridge” between successive class periods.
  2. Improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.
  3. Increase overall creativity of the group.
- **Simulation Tool:**
  1. One of the primary advantages of simulators is that they are able to provide users with practical feedback when designing real world systems.
  2. This allows the designer to determine the correctness and efficiency of a design before the system is actually constructed.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	2	1	1	1	-	-	1	1	1	1	1

CO2: The students will be able to analyze the operation of ARM processor, instruction sets and development tools.

<u>*References:*</u>

- *https://www.youtube.com/\watch?v=7LqPJGnBPMM*
- *https://www.humber.ca*
- *IAR integrated development tool*
- *https://oncourseworkshop.com/self-awareness/one-minute-paper/*
- *Steve Furber, "ARM System –On –Chip architecture," Addision Wesley, 2000.*
- *James L.Antonakos, "An Introduction to the Intel family of Microprocessors," Pearson Education, 1999.*

RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019-2020 (Odd Semester)

Innovative Practices Description for Unit-III

Degree, Semester& Branch: VII Semester B.E. ECE A

Course Code & Title: EC6013 Advanced Microprocessors and Microcontrollers

Name of the Faculty member: Mr.B.Kannan

Name of the Topic: Exception and Interrupts and Caches and Memory Protection Units

Name of the Innovative Practice: Buzz Session and Storyboard Teaching

Date & Duration: 20.08.2019 (30 Minutes) & 26.08.2019 (15 Minutes)

Description:

(i) Buzz Session:

- Buzz sessions are short participative sessions that are deliberately built into a lecture or larger group exercise in order to stimulate discussion and provide student feedback.
- In such sessions, small sub-groups of two to four persons spend a short period (generally no more than five minutes) intensively discussing a topic or topics suggested by the teacher.
- Each sub-group then reports back on its deliberations to the group as a whole, or sometimes combines with another sub-group in order to share their findings and discuss the implications.

(ii) Storyboard Teaching:

- The Storyboards teaching strategy helps students keep track of a narrative's main ideas and supporting details by having them illustrate the story's important scenes.
- Storyboarding can be used when texts are read aloud or when students read independently.
- Checking the thoroughness and accuracy of students' storyboards is an effective way for you to evaluate reading comprehension before moving on to more analytic tasks.

Goals (Learning Outcomes):

- The students will be able to describe the different interrupt handling schemes.
- The students will be able to analyze function and working of cache memory & Memory Protection Units.

Use of appropriate method:

Justification for choosing the following Activities:

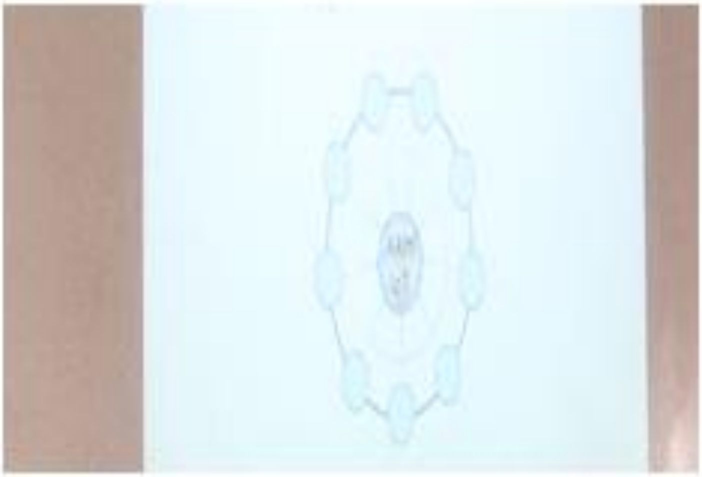
- **Buzz Session:** Buzz sessions are short participative sessions that are deliberately built into a lecture or larger group exercise in order to stimulate discussion and provide student feedback.
- **Storyboard Teaching:** The Storyboards teaching strategy helps students keep track of a narrative's main ideas and supporting details by having them illustrate the story's important scenes.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

- **Buzz Session:** Group created with 3 members with unique topic, asked individual participants to write their own views and discuss with group members at last ask each group to share their points.
- **Storyboard Teaching:** Asked the students to prepare storyboard either in paper or as an image file, with their creation students will explain the concept for the given topics. Few students came with much informative boards.

Innovative Teaching Method Execution

Storyboard Teaching for Caches and Memory Protection Units on 20.08.2019




Buzz Session for Exception and Interrupts of ARM Processor on 26.08.2019


Significance of Results: (Assessment of Effectiveness/Success of the Activity):

- **Buzz Session:** Students delivered the different views on their topics and shared with each groups.
- **Storyboard Teaching:** This technique takes the message among students pictorially and they were interest to learn through storyboard method.

Changes required for future (if required):

- Need to allocate more time to give opportunities to all the students in future.

Reflective Critique:

Challenges:

- **Buzz Session:** Lack of participation by all group members. Some of the students are not attentive towards the accomplishments.
- **Storyboard Teaching:** Took more time to complete the activity and engaging students to make their own storyboards

Benefits:

- **Buzz Session:**
  - Encourage students to become actively involved in a topic.
  - Allow feedback to take place.
  - Short, intense and using trainees own information so there is ownership of the output by trainees.
- **Storyboard Teaching:** Storyboards can help instructors to quickly convey complex ideas to their students

Strengths:

- Can be used to achieve a wide range of objectives, both cognitive and non-cognitive.
- Encourage students to become actively involved in a topic.
- Allow feedback to take place.
- Storyboards can help instructors to quickly convey complex ideas to their students.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	3	2	1	1	1	-	-	1	1	1	1	1

CO3: The students will be able to develop DSP applications using ARM processor.

References:

- *http://www.kstoolkit.org/buzz-groups*
- *www.teachwire.net*
- *https://elearningindustry.com/create-storyboards-for-effective-elearning-8-tips*
- *https://oncourseworkshop.com/self-awareness/one-minute-paper/*
- *ARM System Developer's Guide: Designing and Optimizing System by Andrew N.Sloss*

RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019-2020 (Odd Semester)

Innovative Practices Description for Unit-IV

Degree, Semester& Branch: VII Semester B.E. ECE A

Course Code & Title: EC6013 Advanced Microprocessors and Microcontrollers

Name of the Faculty member: Mr.B.Kannan

Name of the Topic: Operating Modes of 68HC11 and PWM

Name of the Innovative Practice: Zero Minute Speech and Animation

Date & Duration: 07.09.2019 (30 Minutes) & 14.09.2019 (15 Minutes)

Description:

(i) Zero Minute Speech:

- ✪ Zero Minute Speech will leads to express basics of the particular topics by different students.
- ✪ The important topics were allotted to set of students and they will give speech on the topic with doubt clarification.

(ii) Animation:

- ✪ The interactive animation takes less time to learn the students complex things and it makes them enjoy more to learn difficult things, The education and training are higher when the information presented via the computer animation systems than the traditional classroom lectures.
- ✪ Animations help learners understand and remember information and animation is a brilliant and innovative new way to encourage students to learn. Animation teaches using the visual aids, It is a very strong proven way of learning, It brings a topic to life, It gains the attention of the viewer.

Goals (Learning Outcomes):

- ✪ The students will be able to analyse basic mode of 68HC11 processor operation.
- ✪ The students will be able to describe the step by step process of PWM.

Use of appropriate method:

Justification for choosing the following Activities:

- ✪ **Zero Minute Speech:** Leads to express basics of the particular topics by different students.
- ✪ **Animation:** It creates more interest over the topic because PWM is a primary device to control loads with different voltage level, this animation video gives step by step process of PWM

Effective Presentation: (Implementation (Plan & Execution) with Proof):

- ✪ **Zero Minute Speech:** I choose this tool because it does not need a big preparation so I can involve the slow-learners also.
- ✪ **Animation:** Sometimes audio video tool were lengthy session, so I choose the video clips which more essential to the students.

Innovative Teaching Method Execution

Zero Minute Speech for Operating Modes of 68HC11 on 07.09.2019

Animation for PWM on 14.09.2019

Significance of Results: (Assessment of Effectiveness/Success of the Activity):

- **Zero Minute Speech:** Each student will delivered their thoughts briefly related to the topic and clears doubts along with the friends.
- **Animation:** This technique helps to give lively working of PWM to students and they understand the step by step process of PWM.

Changes required for future (if required):

- In future need to plan animation activity for entire period.
- Need to give chance to all students

Reflective Critique:

Challenges:

- **Zero Minute Speech:** Not able to give chance to all students during zero minute speech practice.
- **Animation:** Sometimes there is no audio in animation so faculty member need to explain the step by step working

Benefits:

- **Zero Minute Speech:** Few students share their ideas and information with classmates and some students interacted with other students to clarify the doubts.
- **Animation:**
  - The animation is attractive, It is useful when quickly getting and holding the audience's attention
  - The animation can show the imagined objects in the motion, It is ideal for demonstrating processes
  - Interactive learning with live-action animation keeps learners interested and reinforces skills

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	3	2	1	1	1	-	-	1	1	1	1	1

CO4: The students will be able to explain about MOTOROLA 68HC11 architecture and instruction sets.

References:

- *https://www.youtube.com/watch?v=B_Ysdv1xRbA*
- *Gene.H.Miller, "Micro Computer Engineering," Pearson Education, 2003.*
- *https://www.presentationblogger.com/3-simple-steps-how-to-do-a-speech-on-any-topic-with-zero-preparation/*
- *http://www.icaltefl.com/just-a-minute-speaking-activity*

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019-2020 (Odd Semester)
Innovative Practices Description for Unit-V

Degree, Semester& Branch: VII Semester B.E. ECE A
Course Code & Title: EC6013 Advanced Microprocessors and Microcontrollers
Name of the Faculty member: Mr.B.Kannan
Name of the Topic: Timers and I2C Interfacing
Name of the Innovative Practice: Audio & Video Tools and Exit Slips
Date & Duration: 25.09.2019 (30 Minutes) & 27.09.2019 (10 Minutes)

Description:

(i) Audio & Video Tools:

- Use of audio-visual aids help in maintaining discipline in the class since all the students' attention are focused in learning. This interactive session also develops critical thinking and reasoning that are important components of the teaching-learning process.
- Students learn when they are motivated and curious about something. Traditional verbal instructions can be boring and painful for students. However, use of audio-visual provides intrinsic motivation to students by peaking their curiosity and stimulating their interests in the subjects.

(ii) Exit Slips:

- They provide faculty member with an informal measure of how well students have understood a topic or lesson.
- They help students reflect on what they have learned.
- They allow students to express what or how they are thinking about new information.
- They teach students to think critically.
- At the end of your lesson ask students to respond to a question or prompt.
  - Prompts that document learning:
    Example: Write one thing you learned today.
    Example: Discuss how today lesson could be uses in the real world.
  - Prompts that emphasize the process of learning:
    Example: I didn't understand…
    Example: Write one question you have about today's lesson.
  - Prompts to evaluate the effectiveness of instruction:
    Example: Did you enjoy working in small groups today?

Goals (Learning Outcomes):

- The students will be able to describe the function of timer.
- The students will be able to elaborate operation of I2C bus.

Use of appropriate method:

Justification for choosing the following Activities:

- **Audio & Video Tools:** According to the Webster dictionary, audio-visual aids are defined as "training or educational materials directed at both the senses of hearing and the sense. Audio-visual provides opportunities for effective communication between teacher and students in learning.
- **Exit Slips:** It allow the faculty member to measure understanding level of the students.

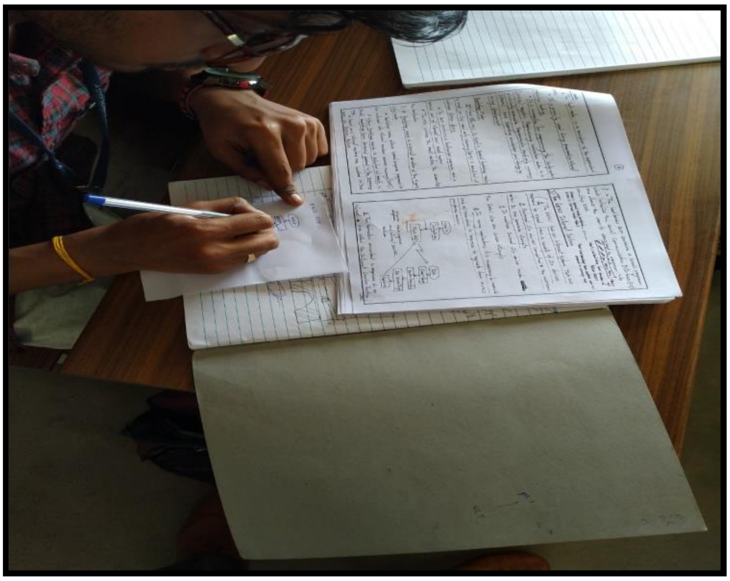
Effective Presentation: (Implementation (Plan & Execution) with Proof):

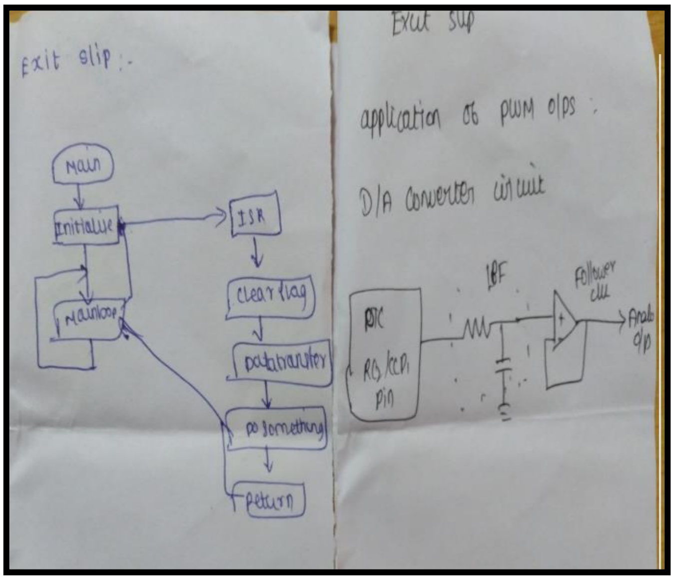
- **Audio & Video Tools:** Sometimes audio video tool were lengthy session, so I choose the video clips which more essential to the students.
- **Exit Slips:** It allow the faculty member to measure understanding level of the students. I gave brief introduction about how to write the exit slip with more informative.

Innovative Teaching Method Execution

Audio & Video Tools for Timer on 25.09.2019

Exit Slip for I2C Interface on 27.09.2019





Significance of Results: (Assessment of Effectiveness/Success of the Activity):

- **Audio & Video Tools:** With audio & video presentation the students got different types of procedures and techniques.
- **Exit Slips:** With in a time they can recall the specific topic and put it in the exist slip, which leads to remember the topic for long time.

Changes required for future (if required):

- In future exit slip activity should conduct through Google forms.
- Need to select better quality videos in futures.

Reflective Critique:

Challenges:

- **Audio & Video Tools:** Sometimes audio video tool were lengthy session.
- **Exit Slips:** Not all students actively participated.

Benefits:

- **Audio & Video Tools:** Audio video tool gives different teaching experience to the students with some animation.
- **Exit Slips:**
  - Prompts that document learning.
  - Prompts that emphasize the process of learning.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	3	2	1	1	1	-	-	1	1	1	1	1

CO5: The students will be able to elaborate the operation of PIC microcontroller and instruction sets.

References:

- *https://www.sdera.wa.edu.au*
- *https://technologyadvice.com/blog/marketing/brainstorming-activities-inspire-content-marketing-team/*
- *John.B.Peatman, "Design with PIC Microcontroller," Prentice Hall, 1997.*
- *https://www.wrike.com/blog/techniques-effective-brainstorming/*
- *https://www.readingrockets.org/strategies/exit_slips*
- *http://www.readwritethink.org/professional-development/strategy-guides/exit-slips-30760.html*